ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2025

Assignment 6 - Due date 02/27/25

Student Name

Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A06_Sp25.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: "ggplot2", "forecast", "tseries" and "sarima". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
#Load/install required package here
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(ggplot2)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(Kendall)
library(tseries)
library(outliers)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr   1.1.4     v stringr 1.5.1
## v forcats 1.0.0     v tibble  3.2.1
## v purrr   1.0.2     v tidyr   1.3.1
## v readr   2.1.5

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

This assignment has general questions about ARIMA Models.

Q1

Describe the important characteristics of the sample autocorrelation function (ACF) plot and the partial sample autocorrelation function (PACF) plot for the following models:

- AR(2)

  Answer: The ACF will likely show a slow decay. The PACF will have a cut off at lag 2.

- MA(1)

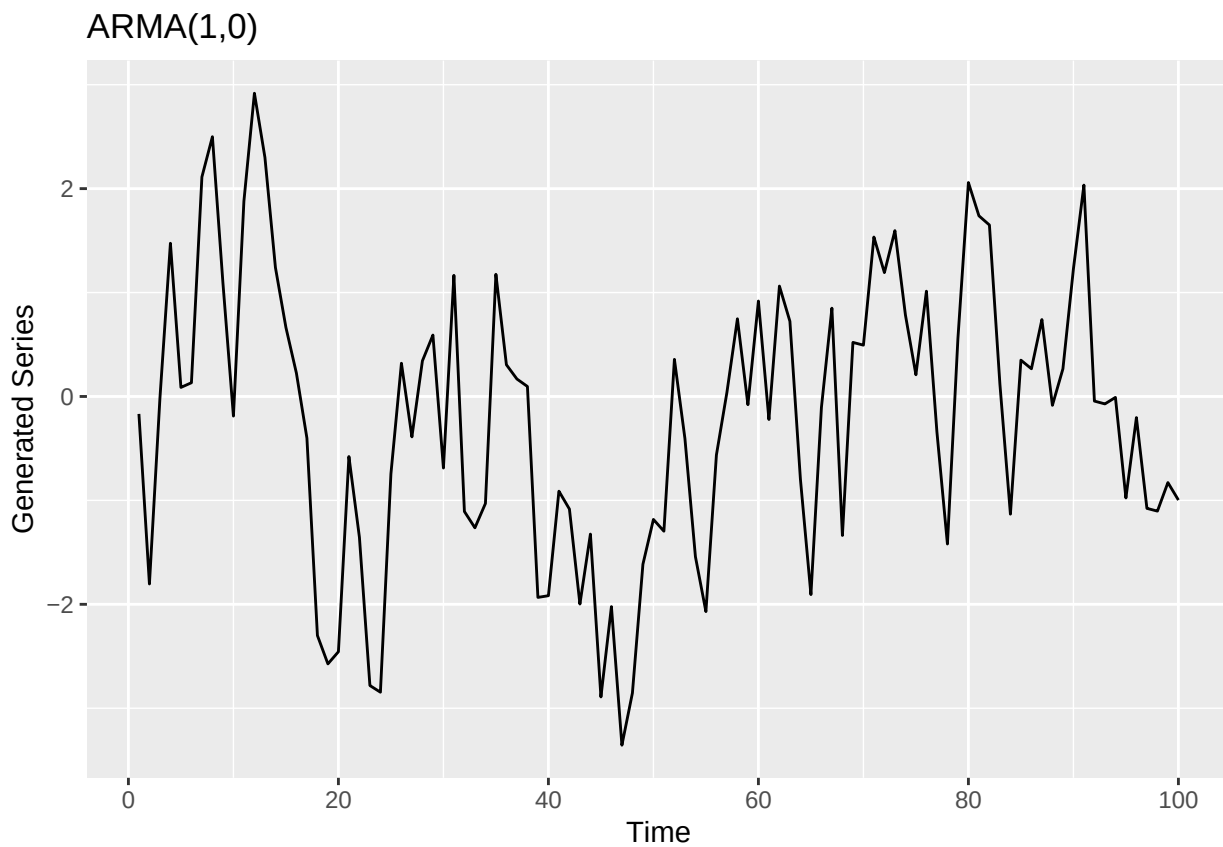
  Answer: The ACF will have a significant value at lag 1. After that the values may oscillate but will not be significant. The PACF may show a slow decay.

Q2

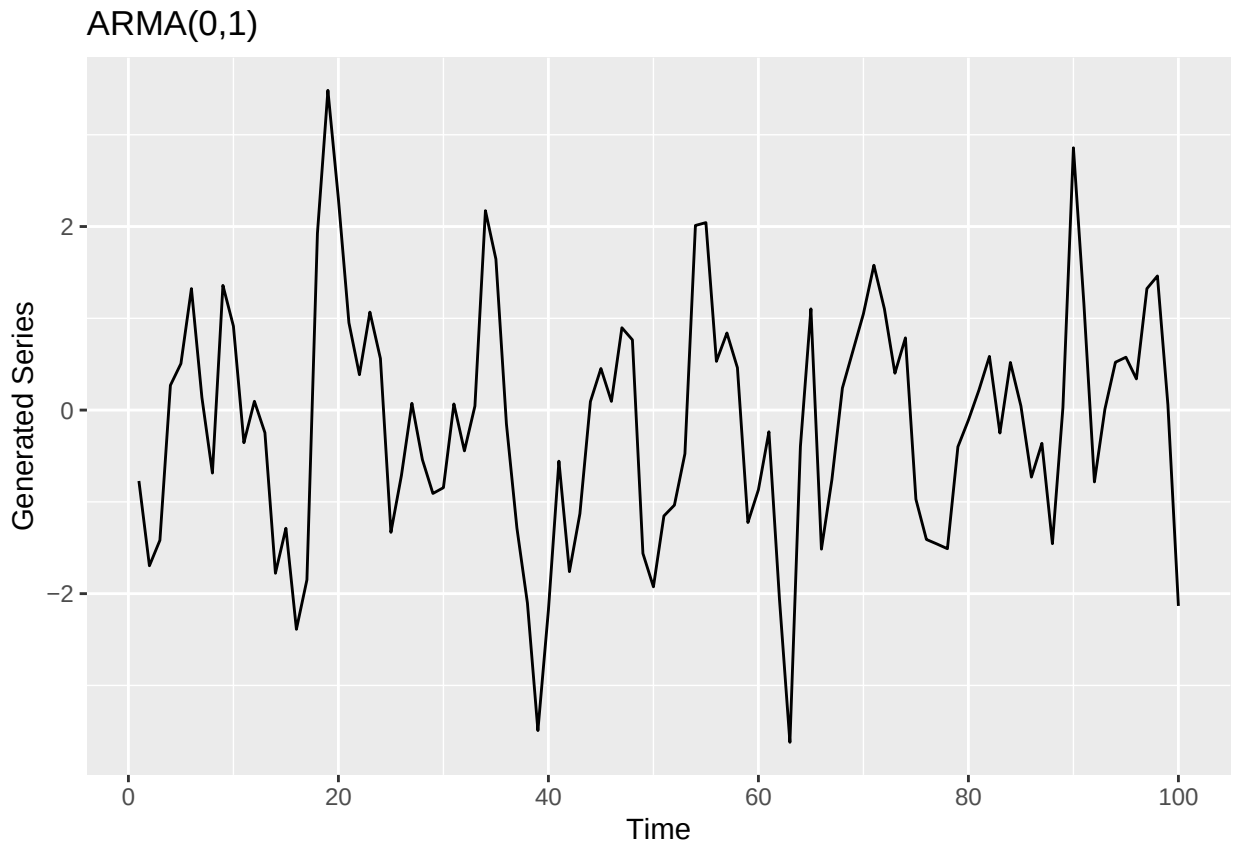
Recall that the non-seasonal ARIMA is described by three parameters ARIMA(p, d, q) where p is the order of the autoregressive component, d is the number of times the series need to be differenced to obtain stationarity and q is the order of the moving average component. If we don't need to difference the series, we don't need to specify the "I" part and we can use the short version, i.e., the ARMA(p, q).

(a) Consider three models: ARMA(1,0), ARMA(0,1) and ARMA(1,1) with parameters $\phi = 0.6$ and $\theta = 0.9$. The ϕ refers to the AR coefficient and the θ refers to the MA coefficient. Use the `arima.sim()` function in R to generate $n = 100$ observations from each of these three models. Then, using `autoplot()` plot the generated series in three separate graphs.

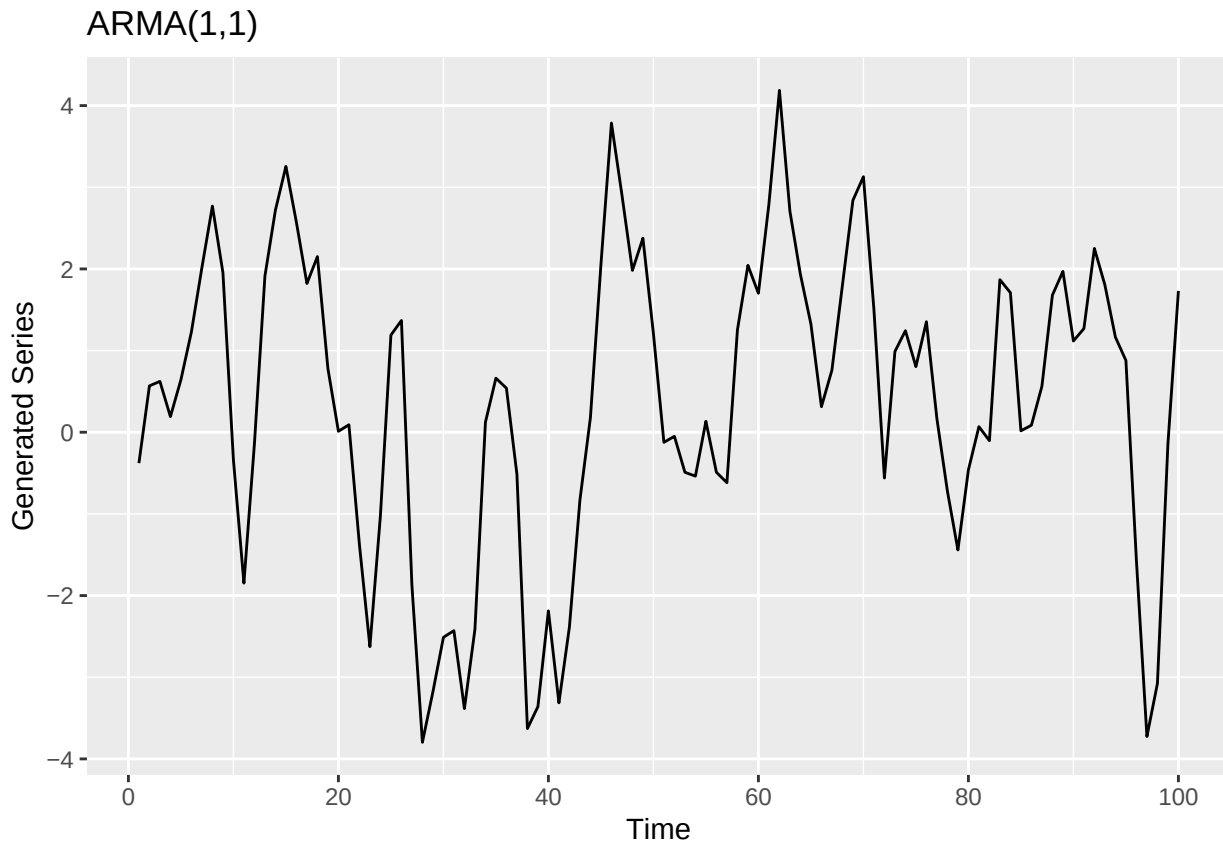
```r
ARMA10 <- arima.sim(list(order=c(1,0,0),ar=c(0.6)),n=100)
ARMA01 <- arima.sim(list(order=c(0,0,1),ma=c(0.9)),n=100)
ARMA11 <- arima.sim(list(order=c(1,0,1),ar=c(0.6),ma=c(0.9)),n=100)

autoplot(ARMA10)+
  ggtitle("ARMA(1,0)")+
  ylab("Generated Series")
```




```r
autoplot(ARMA01)+
  ggtitle("ARMA(0,1)")+
  ylab("Generated Series")
```



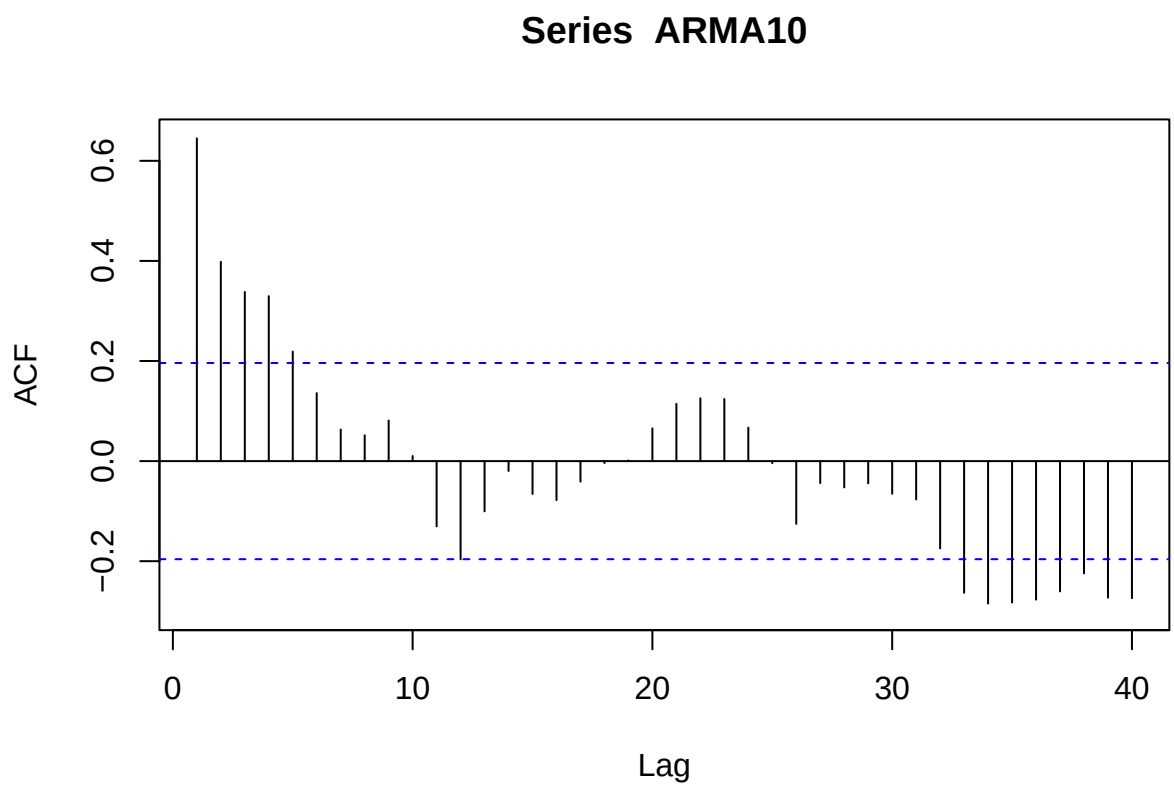

```
autoplot(ARMA11)+
  ggtitle("ARMA(1,1)")+
  ylab("Generated Series")
```



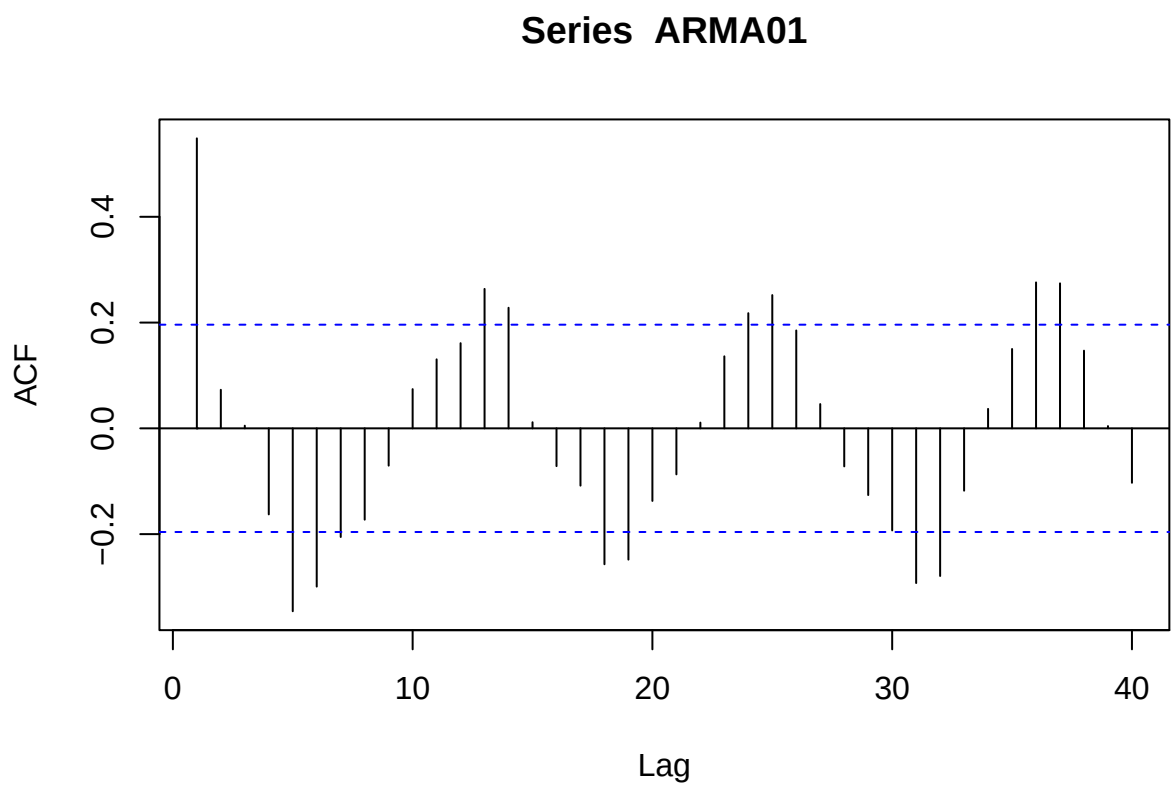
(b) Plot the sample ACF for each of these models in one window to facilitate comparison (Hint: use `cowplot::plot_grid()`).

```
plot_grid(
  autoplot(Acf(ARMA10, lag.max=40),main = "ARMA(1,0)"),
  autoplot(Acf(ARMA01, lag.max=40), main = "ARMA(0,1)"),
  autoplot(Acf(ARMA11, lag.max=40),main = "ARMA(1,1)")
)
```

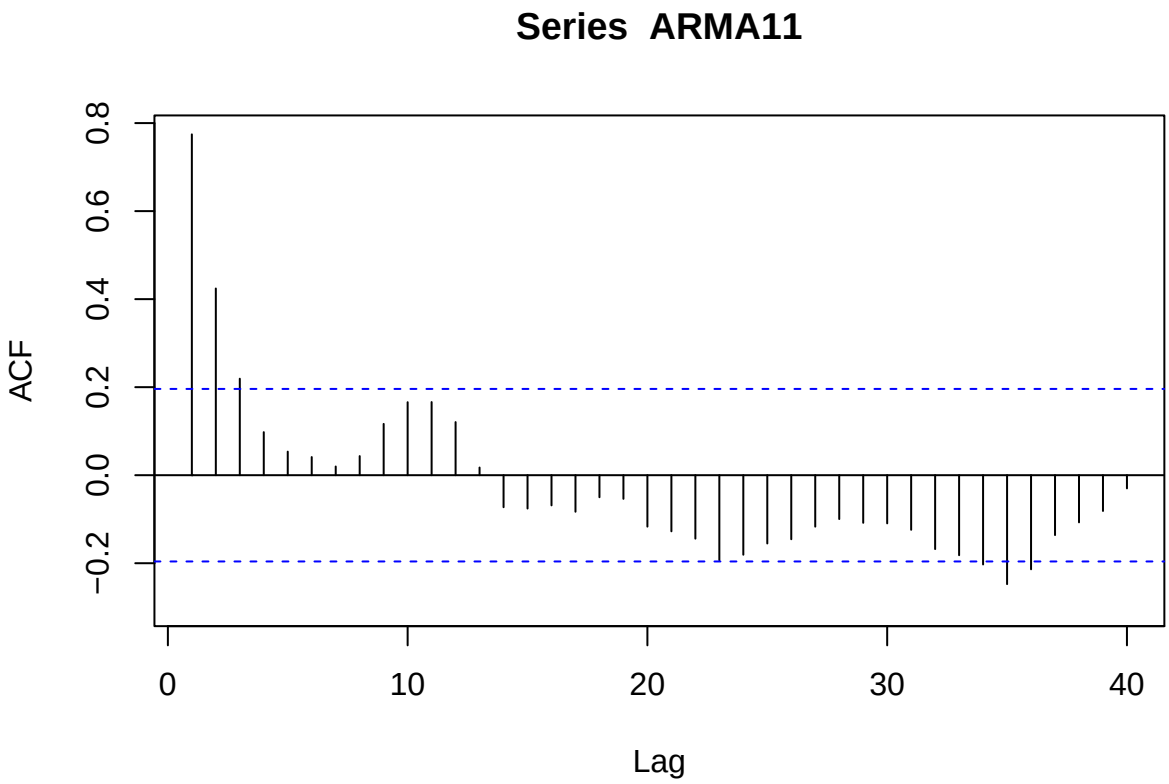
```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: ‘main‘
```

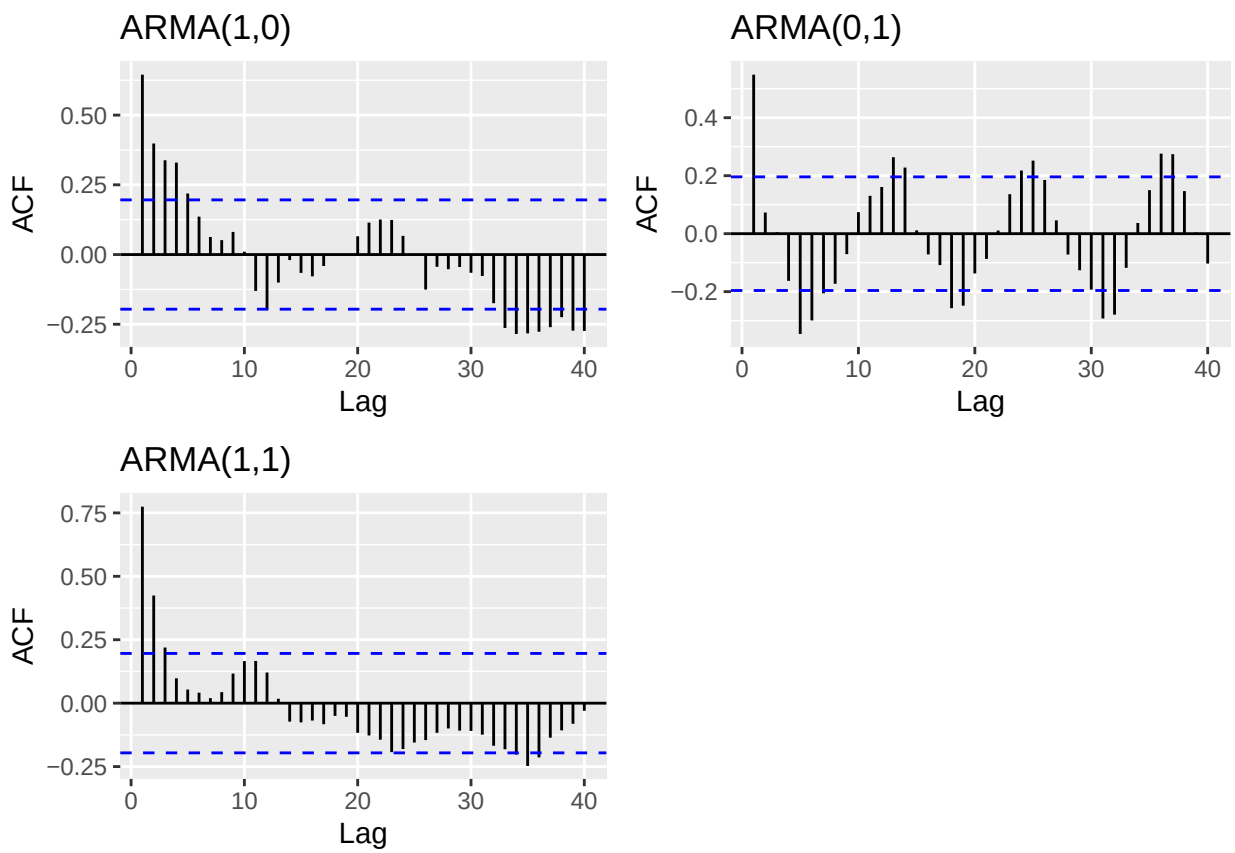



```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```




```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```

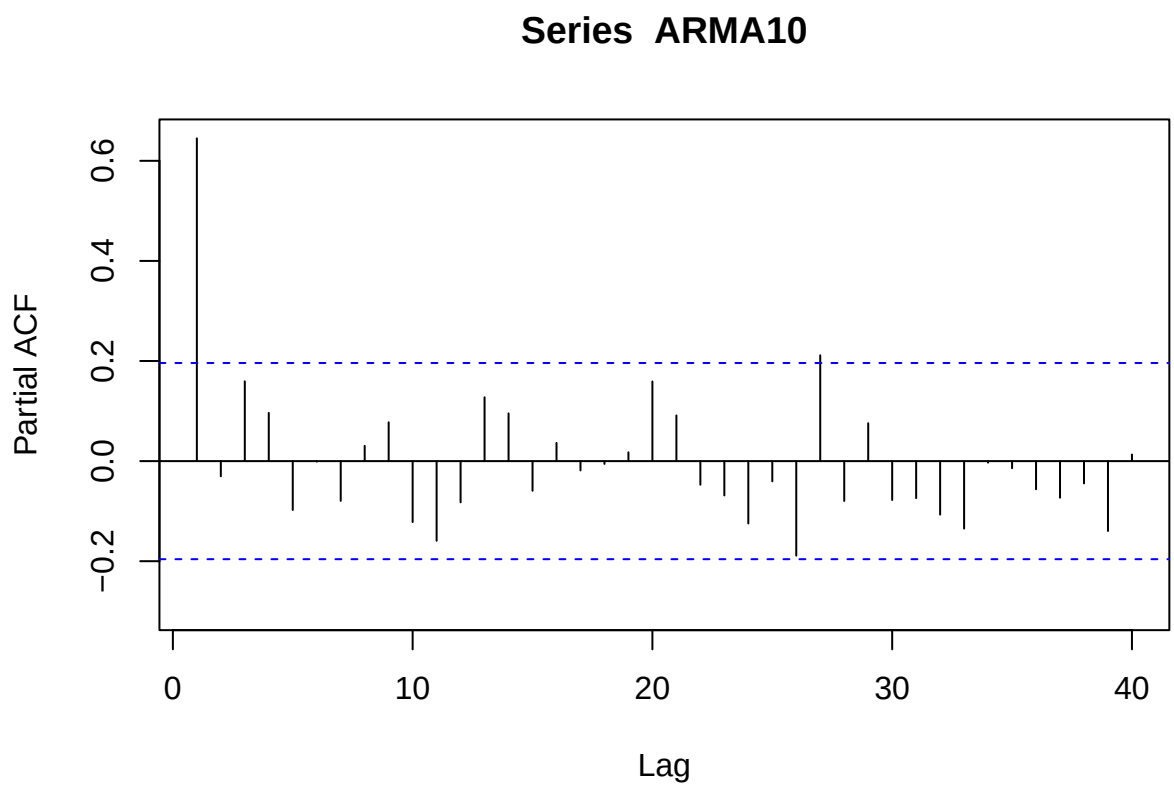

Series ARMA11
ACF
0.8
0.6
0.4
0.2
0.0
−0.2
0
10
20
30
40
Lag



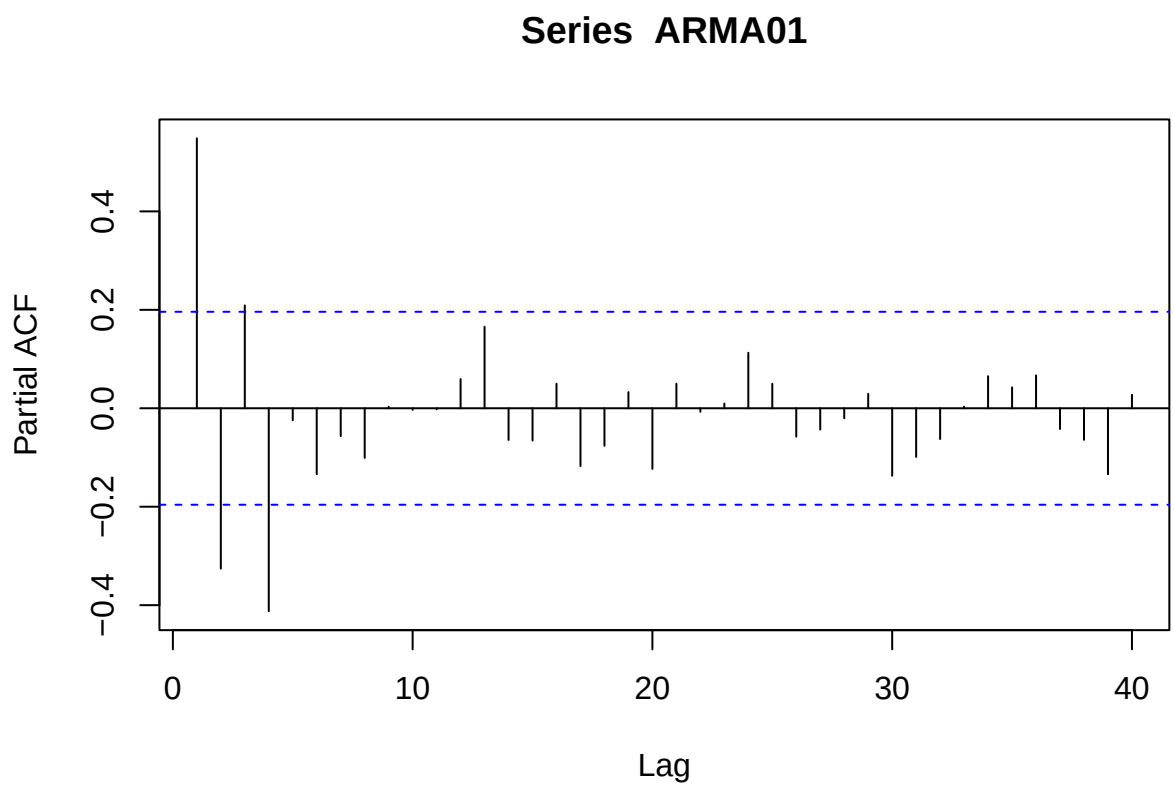
(c) Plot the sample PACF for each of these models in one window to facilitate comparison.

```
plot_grid(
  autoplot(Pacf(ARMA10, lag.max=40),main = "ARMA(1,0)"),
  autoplot(Pacf(ARMA01, lag.max=40), main = "ARMA(0,1)"),
  autoplot(Pacf(ARMA11, lag.max=40),main = "ARMA(1,1)")
)
```

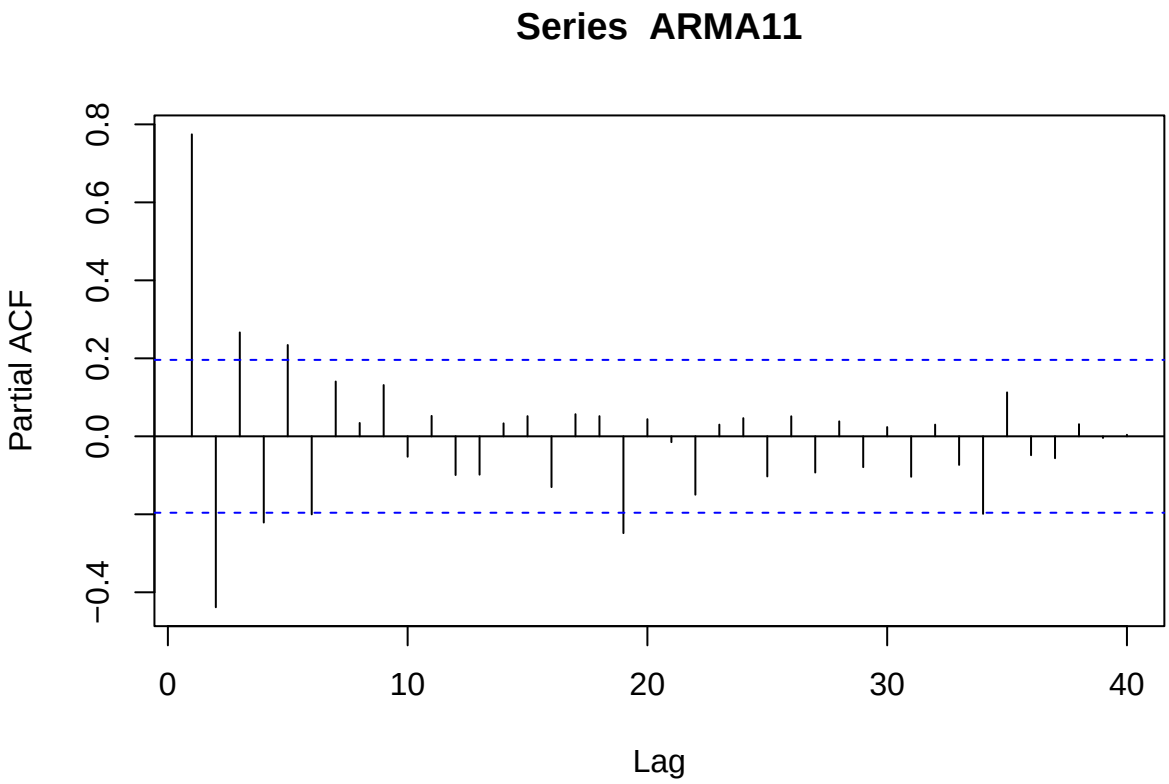
```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```

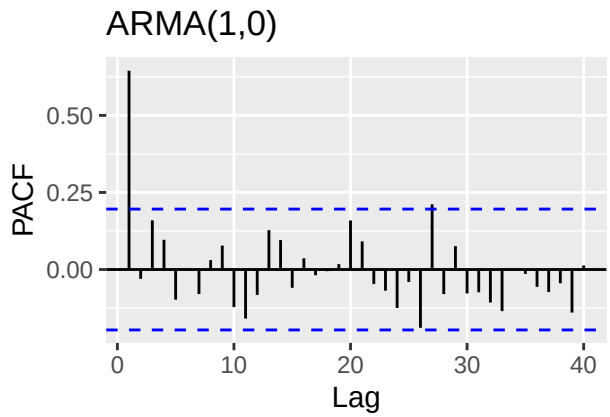


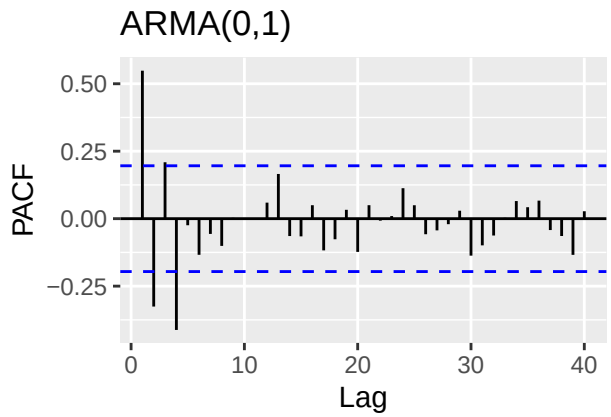

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```

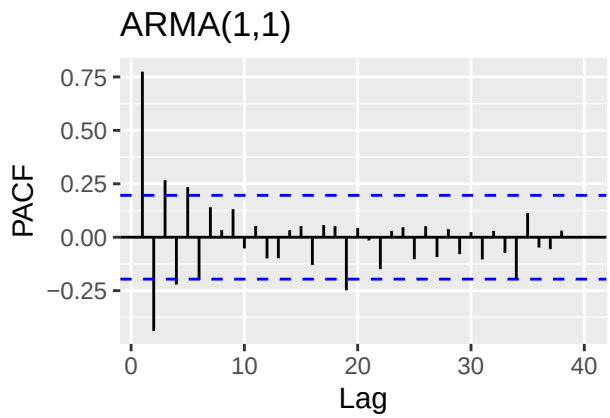



```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```


Series ARMA11
Partial ACF
0.8
0.6
0.4
0.2
0.0
−0.4
0
10
20
30
40
Lag







(d) Look at the ACFs and PACFs. Imagine you had these plots for a data set and you were asked to identify the model, i.e., is it AR, MA or ARMA and the order of each component. Would you be able identify them correctly? Explain your answer.

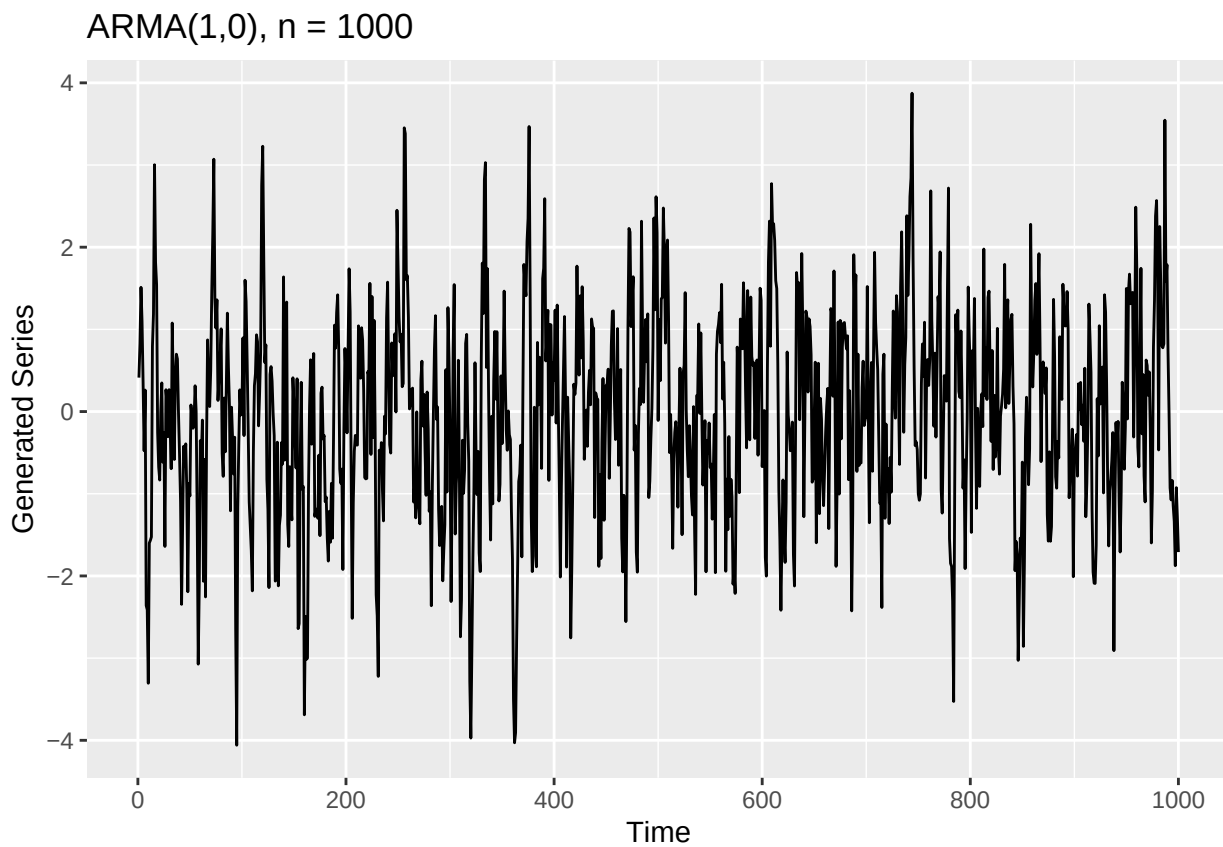
Answer:The models with either AR or MA components are identifiable. For ARMA(1,0), the PACF shows a significant spike at lag 1 and then the rest of the values are within the blue lines. For purely AR models, the PACF tells us what order the model is based on the cut off value. Here, the cutoff and the AR order are 1. The ACF for this model does not have a very obvious slow decay, but there is still a slow decay with significant values still occuring around lag 8. For ARMA(0,1), the ACF has a distinct cut off at lag 1. For purely MA models, the ACF tells us what order the model is based on the cut off value. Here, the cutoff and the MA order are 1. The PACF has a slower decay, which is typical of an MA model. The ARMA(1,1) does not have clear cut offs for the ACF or PACf, which tells us that this is not a purely MA or AR model. Beyond that, it would be hard to identify the order.

(e) Compare the PACF values R computed with the values you provided for the lag 1 correlation coefficient, i.e., does $\phi = 0.6$ match what you see on PACF for ARMA(1,0), and ARMA(1,1)? Should they match?

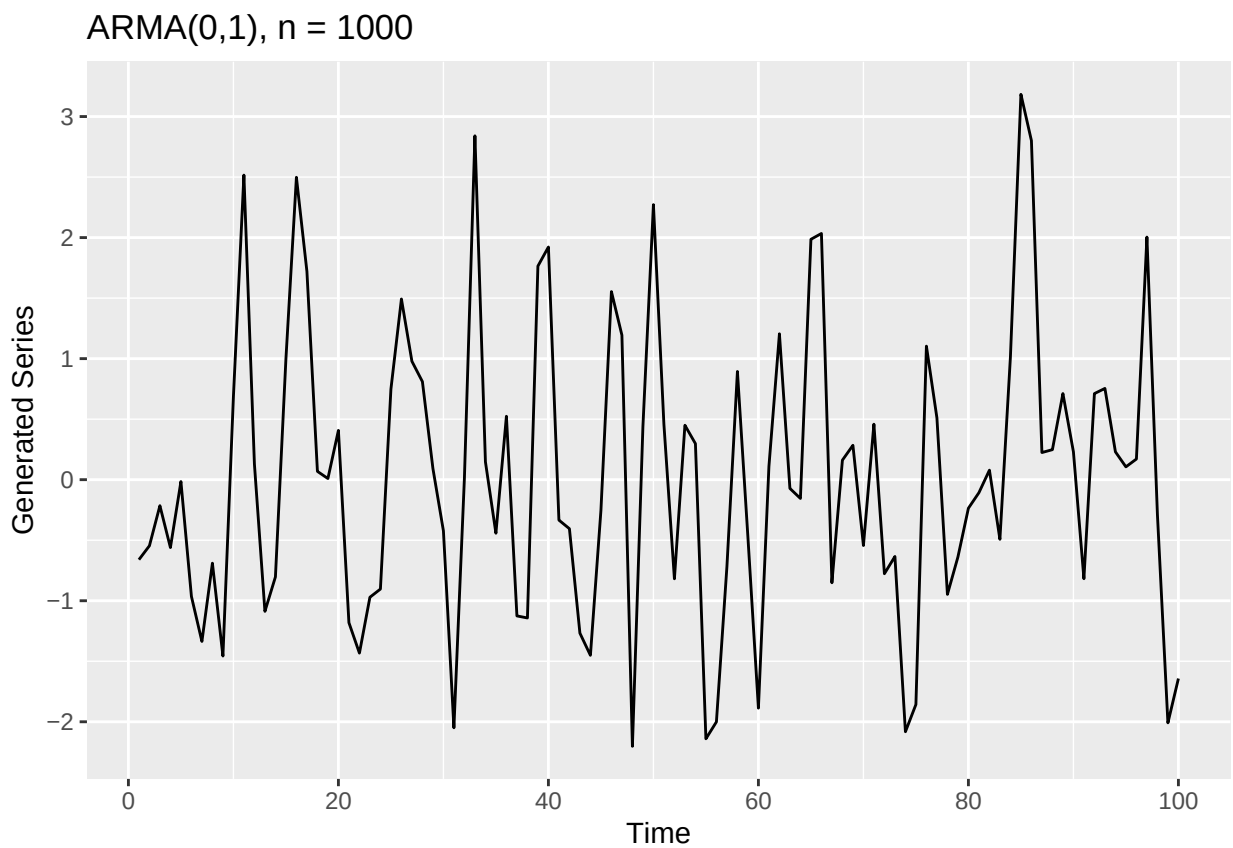
Answer: The PACF values for ARMA (1,0) and ARMA(1,1) do not match. The ARMA(1,0) PACF lag 1 is close to its coefficient (0.6), but not exactly because the AR equation also includes an error term. The ARMA(1,1) PACF at lag 1 is not near 0.6 (it's closer to 0.8). This model does incorporate the AR component with the 0.6 coefficient and error term, but also includes the MA component and it's coefficient (0.9).

(f) Increase number of observations to $n = 1000$ and repeat parts (b)-(e).

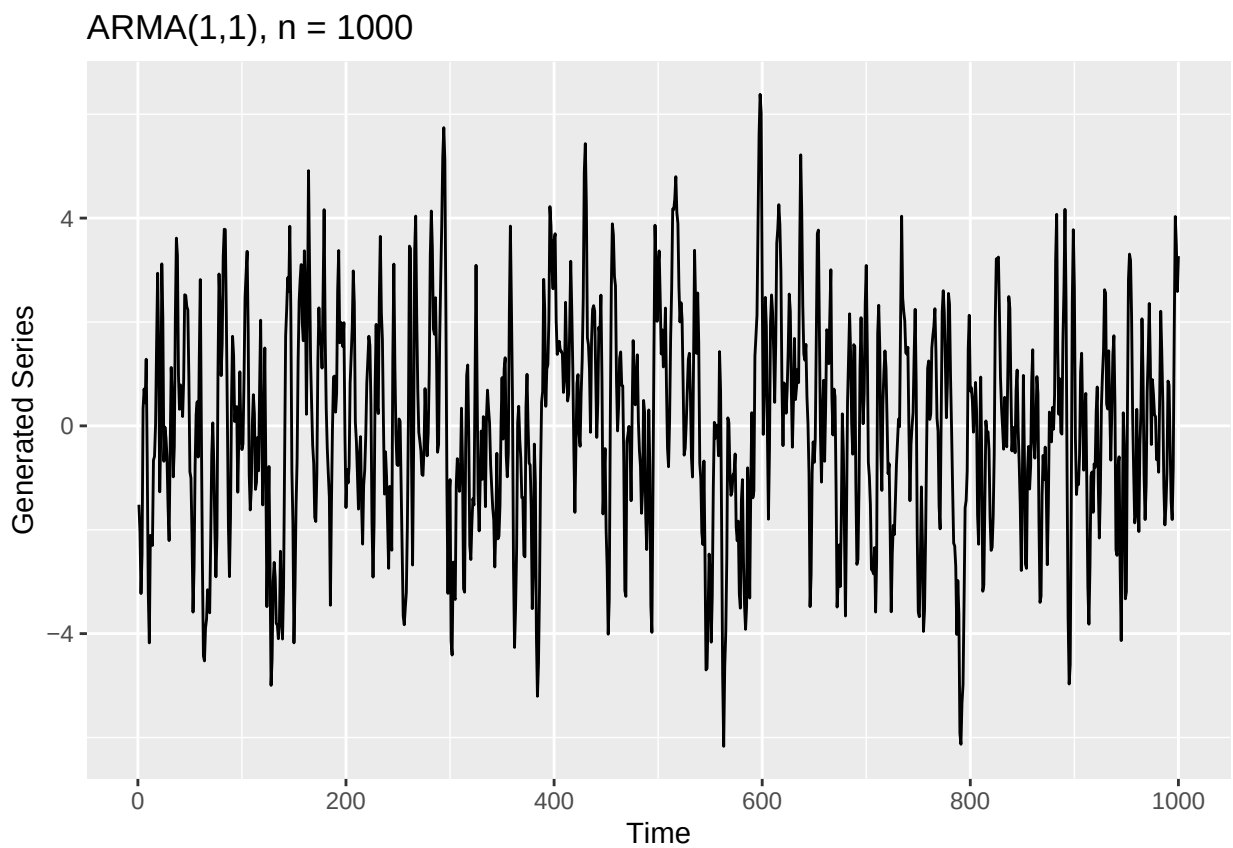
```r
ARMA10.b <- arima.sim(list(order=c(1,0,0),ar=c(0.6)),n=1000)
ARMA01.b <- arima.sim(list(order=c(0,0,1),ma=c(0.9)),n=100)
ARMA11.b <- arima.sim(list(order=c(1,0,1),ar=c(0.6),ma=c(0.9)),n=1000)

autoplot(ARMA10.b)+
  ggtitle("ARMA(1,0), n = 1000")+
  ylab("Generated Series")
```




```r
autoplot(ARMA01.b)+
  ggtitle("ARMA(0,1), n = 1000")+
  ylab("Generated Series")
```

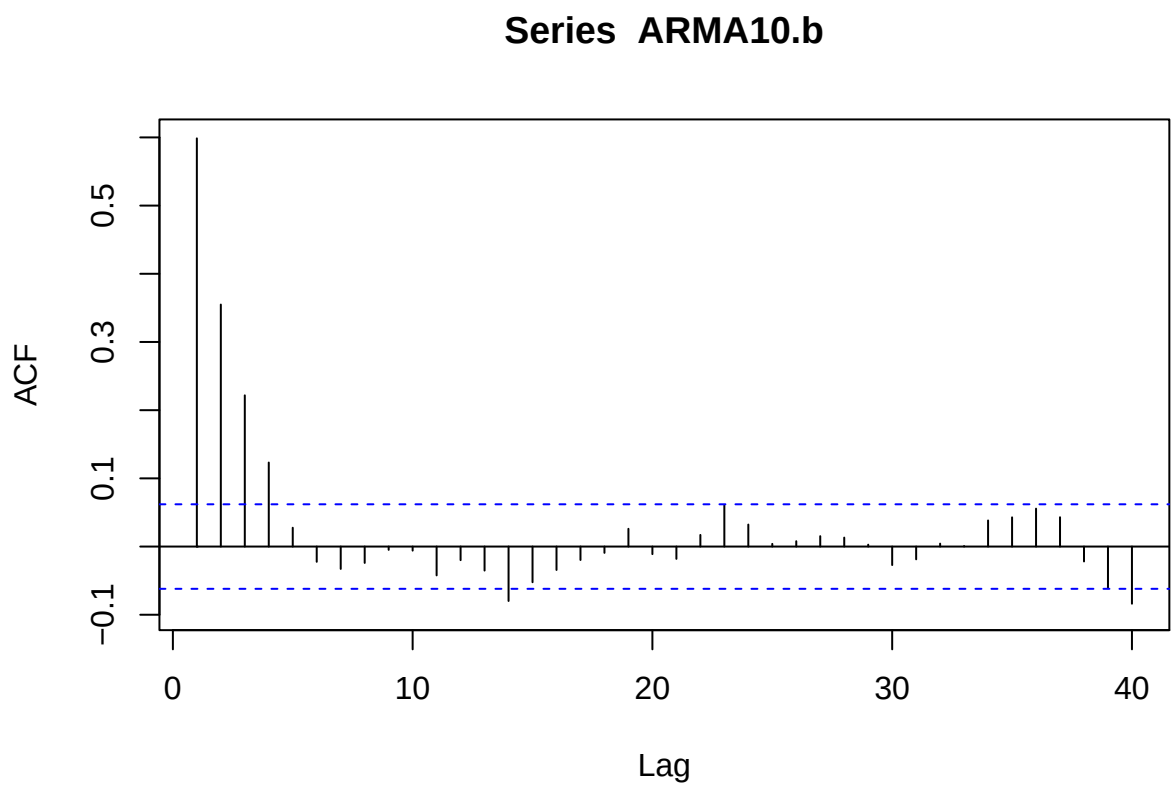



```
autoplot(ARMA11.b)+
  ggtitle("ARMA(1,1), n = 1000")+
  ylab("Generated Series")
```

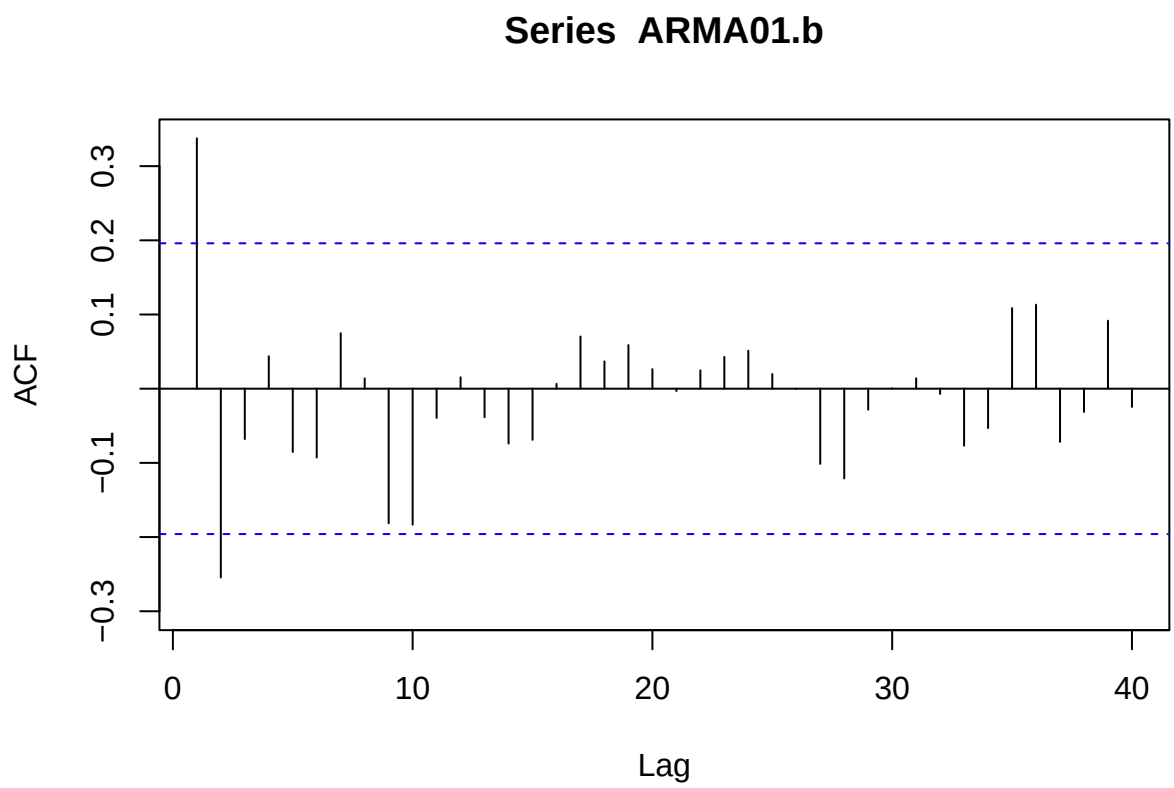



```r
print(plot_grid(
  autoplot(Acf(ARMA10.b, lag.max=40),main = "ARMA(1,0)"),
  autoplot(Acf(ARMA01.b, lag.max=40), main = "ARMA(0,1)"),
  autoplot(Acf(ARMA11.b, lag.max=40),main = "ARMA(1,1)")
))
```

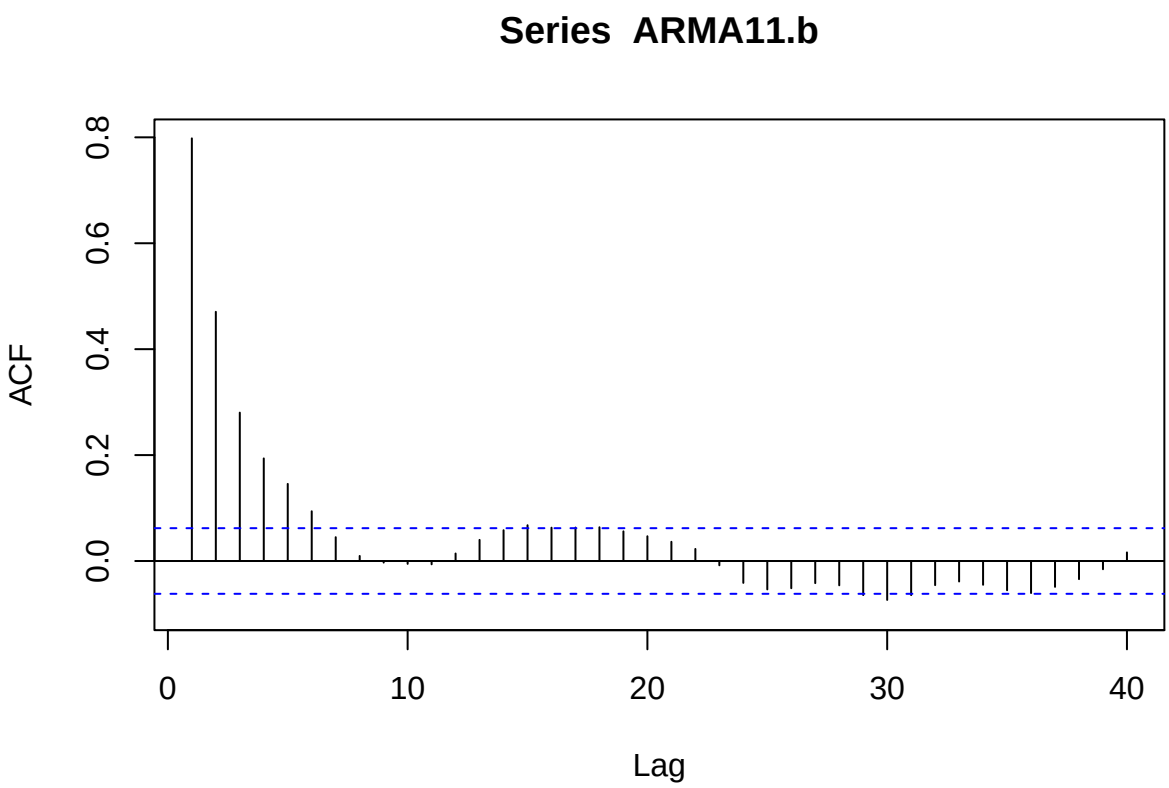
```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```

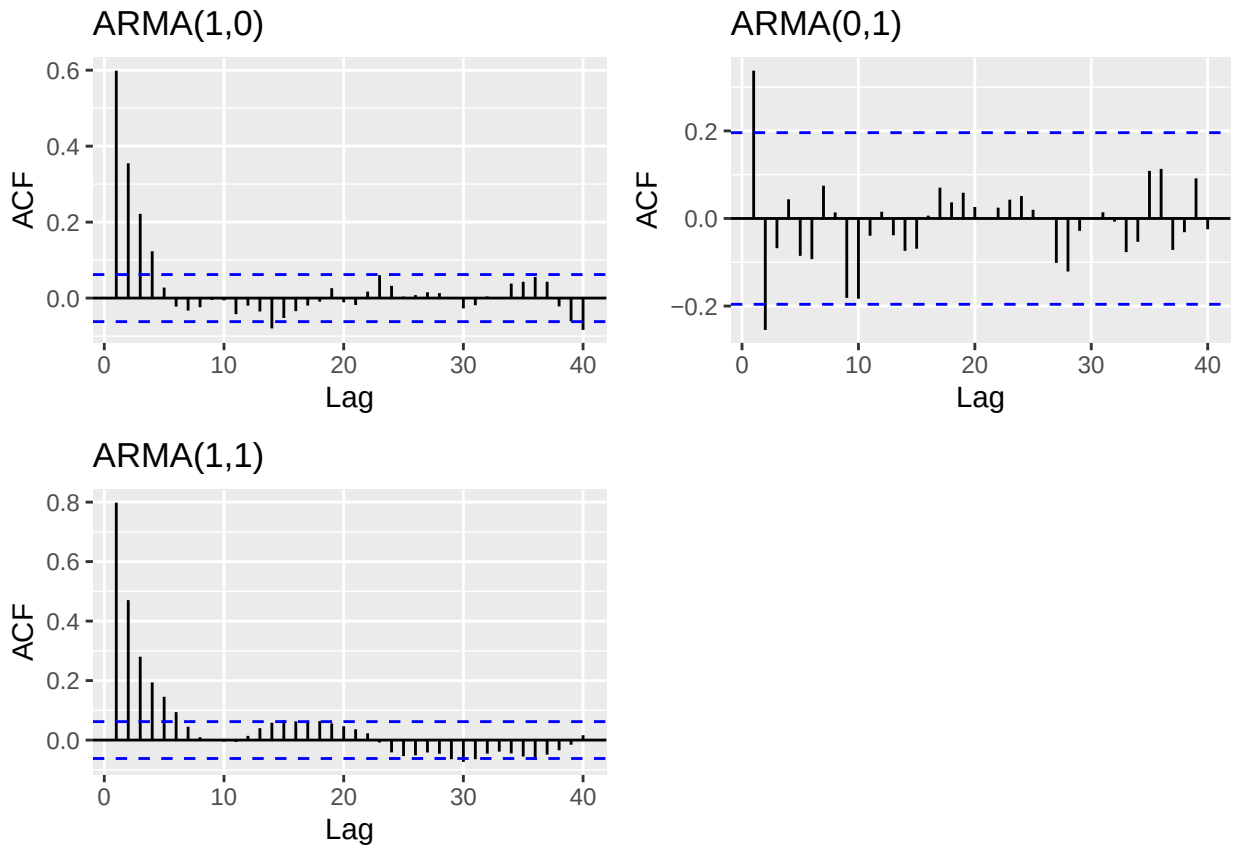



```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```



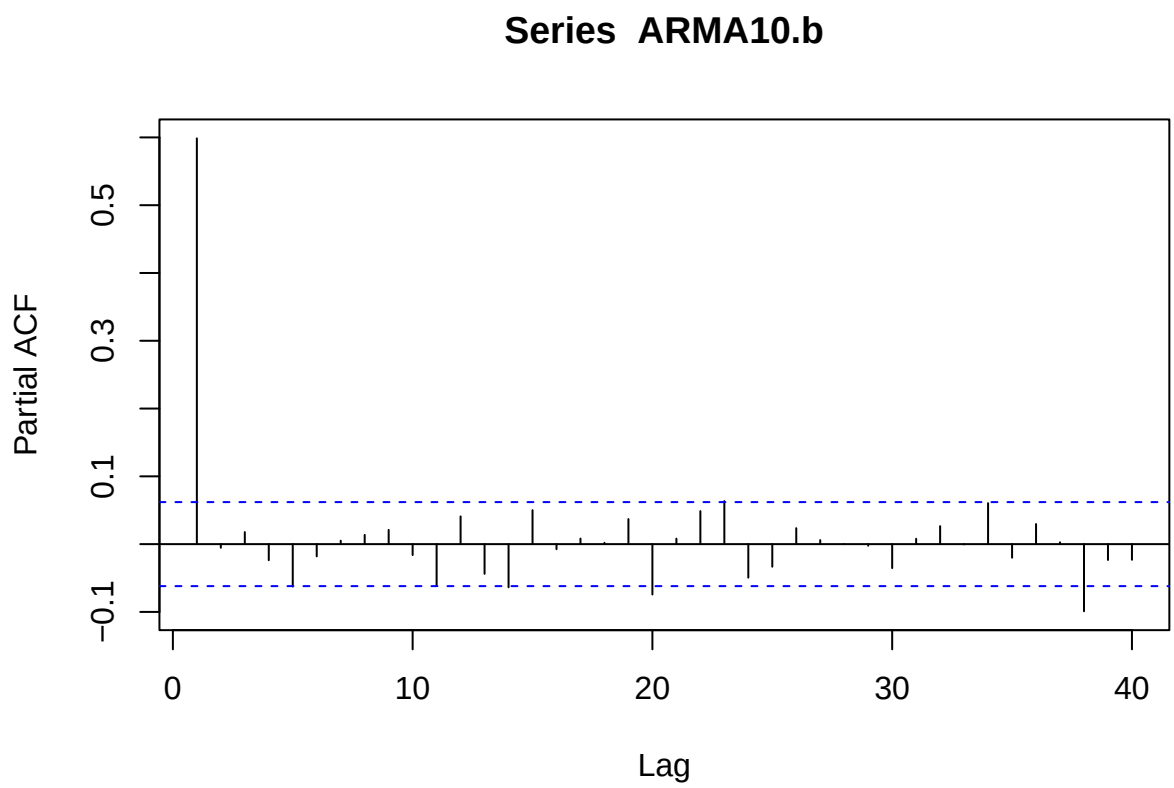

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```


Series ARMA11.b
ACF
0.0
0.2
0.4
0.6
0.8
0
10
20
30
40
Lag

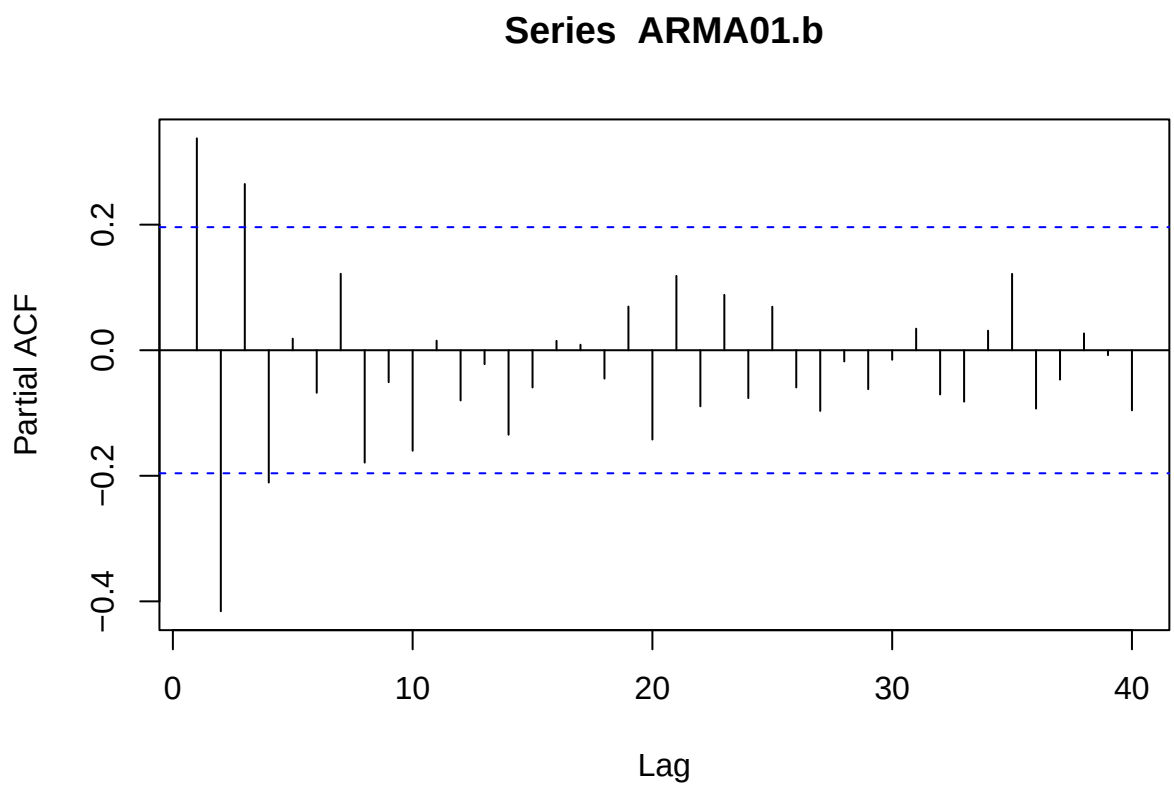



```r
plot_grid(
  autoplot(Pacf(ARMA10.b, lag.max=40),main = "ARMA(1,0)"),
  autoplot(Pacf(ARMA01.b, lag.max=40), main = "ARMA(0,1)"),
  autoplot(Pacf(ARMA11.b, lag.max=40),main = "ARMA(1,1)")
)
```

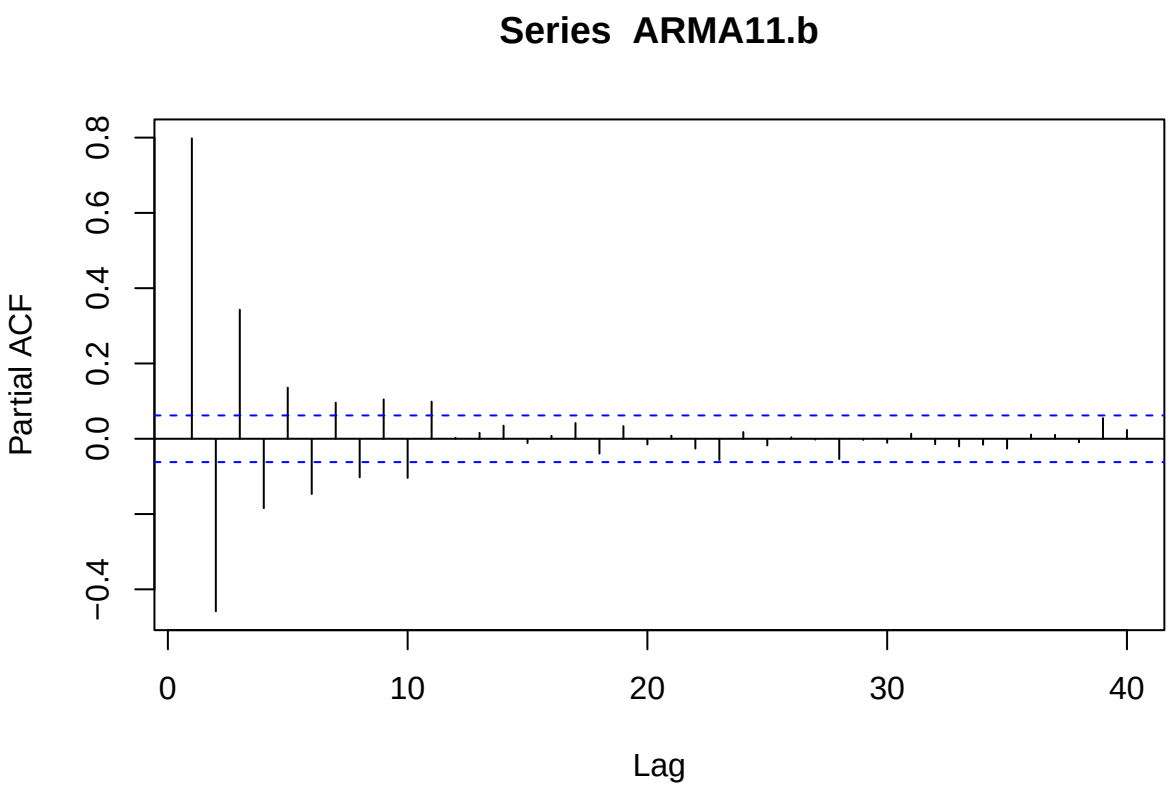
```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```

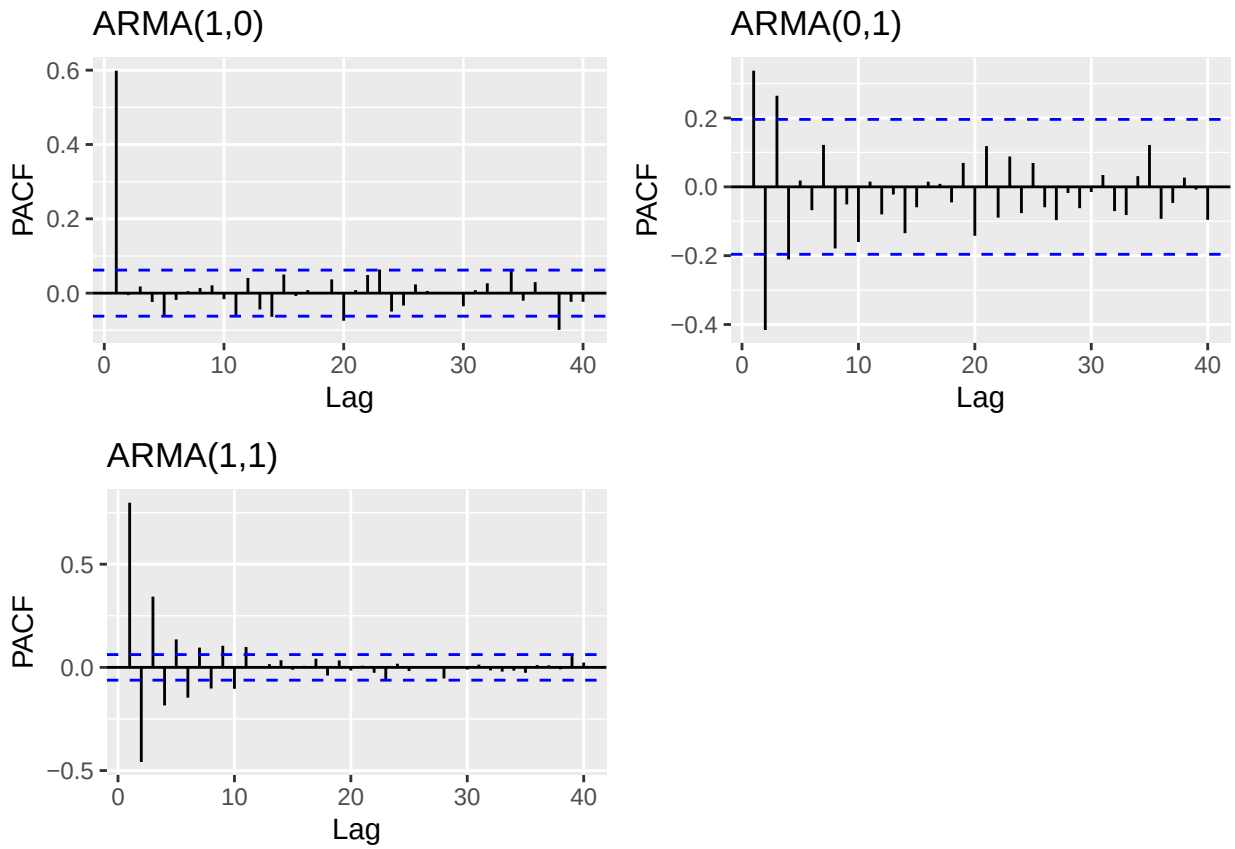



```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```

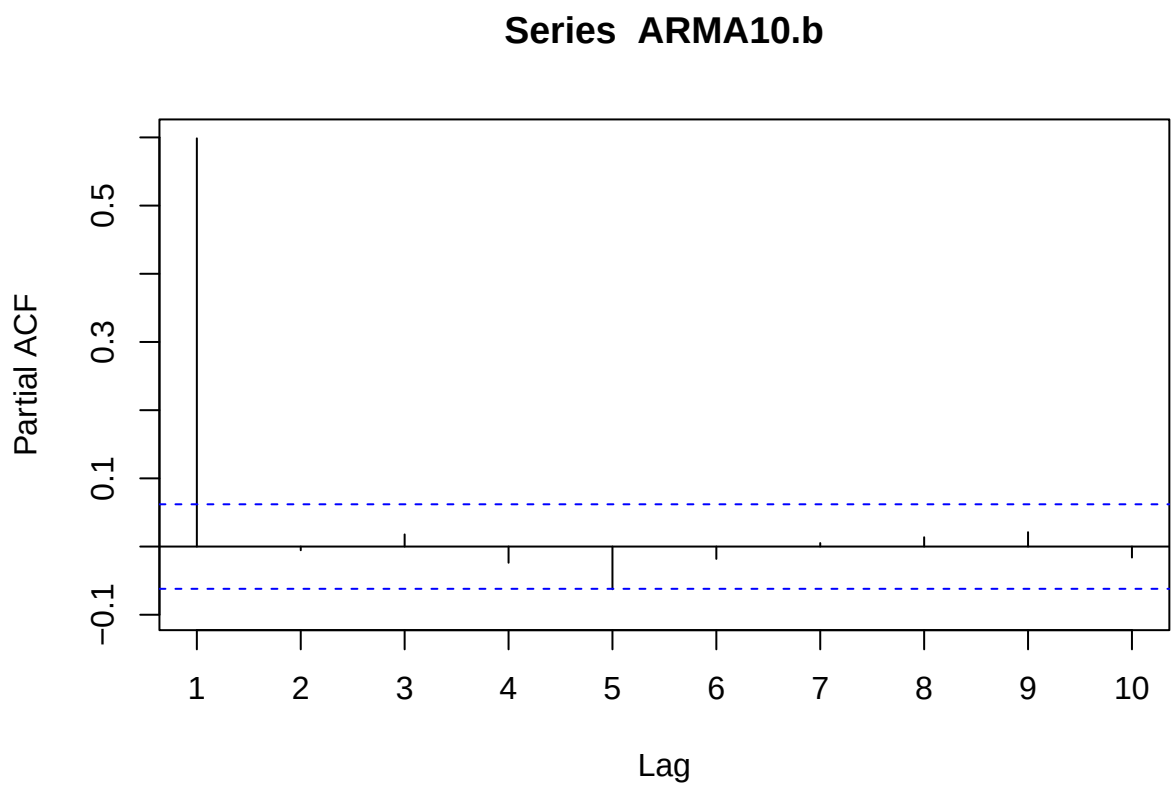



```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `main`
```


Series ARMA11.b
Partial ACF
0.8
0.6
0.4
0.2
0.0
−0.4
0
10
20
30
40
Lag

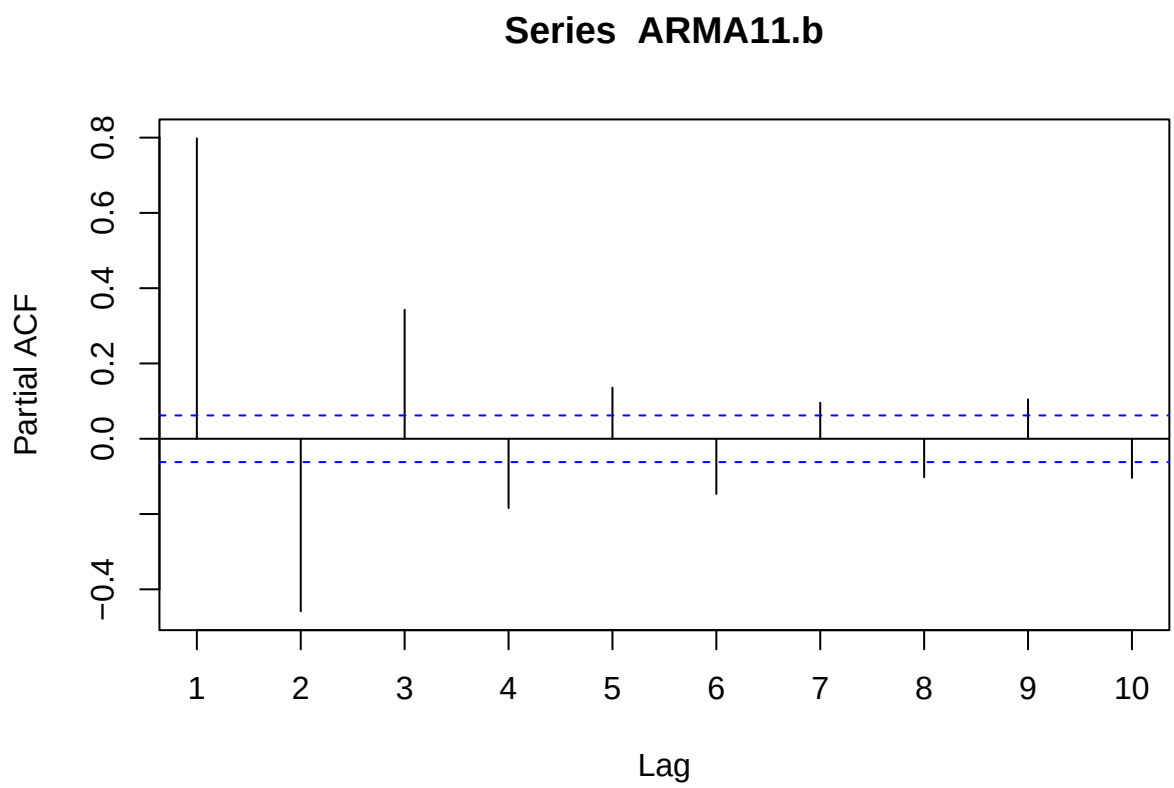



```r
print(Pacf(ARMA10.b, lag.max = 10))
```




```
##
## Partial autocorrelations of series 'ARMA10.b', by lag
##
##      1      2      3      4      5      6      7      8      9     10
##  0.599 -0.005  0.018 -0.024 -0.063 -0.018  0.005  0.014  0.021 -0.016
```

```r
print(Pacf(ARMA11.b, lag.max = 10))
```




```
##
## Partial autocorrelations of series 'ARMA11.b', by lag
##
##      1      2      3      4      5      6      7      8      9     10
##  0.798 -0.458  0.343 -0.184  0.136 -0.147  0.096 -0.102  0.104 -0.104
```

The PACF and ACF for the most part would help predict the type of model (PACF for ARMA(1,0) has a sharp cut off at lag one and there is a cutoff at lag one for the ACF ARMA (0,1). However, the ACF for ARMA(0,1) has a few spikes that might make me think this would have AR and MA components. The ARMA(1,1) has a slow decay for both the PACF and ACF ## Q3

Consider the ARIMA model $y_t = 0.7 * y_{t-1} - 0.25 * y_{t-12} + a_t - 0.1 * a_{t-1}$

(a) Identify the model using the notation ARIMA$(p, d, q)(P, D, Q)_s$, i.e., identify the integers p, d, q, P, D, Q, s (if possible) from the equation.

(b) Also from the equation what are the values of the parameters, i.e., model coefficients.

Q4

Simulate a seasonal ARIMA$(0, 1) \times (1, 0)_{12}$ model with $\phi = 0.8$ and $\theta = 0.5$ using the `sim_sarima()` function from package `sarima`. The 12 after the bracket tells you that $s = 12$, i.e., the seasonal lag is 12, suggesting monthly data whose behavior is repeated every 12 months. You can generate as many observations as you like. Note the Integrated part was omitted. It means the series do not need differencing, therefore $d = D = 0$. Plot the generated series using `autoplot()`. Does it look seasonal?

Q5

Plot ACF and PACF of the simulated series in Q4. Comment if the plots are well representing the model you simulated, i.e., would you be able to identify the order of both non-seasonal and seasonal components from the plots? Explain.